

Extreme URLLC and Tactile Internet for XR, Holographic, and Metaverse Applications

Gihan Kanishka, Ginura Binath

Abstract—Future communication networks must support new immersive digital applications. Extended reality, holographic communication, and the metaverse are some of the examples. These applications require extremely fast and reliable communication. Traditional communication systems focus mainly on high data speed. However, immersive systems require low delay and stable network performance. Extreme Ultra-Reliable Low-Latency Communication (xURLLC) and the Tactile Internet are useful technologies for supporting these requirements. This paper provides a survey of research related to immersive communication systems. It explains the basic concepts of extreme URLLC and the tactile internet. It also explains the communication requirements of XR, holographic communication, and metaverse systems. The paper discusses enabling technologies such as edge computing, artificial intelligence, and network slicing. Research approaches from recent studies are compared. Limitations and research challenges are identified. Finally, future research directions for immersive communication systems are discussed. This paper studies these technologies and explains how they support immersive communication systems.

I. INTRODUCTION

Wireless communication systems have evolved significantly over the past few decades. The first generation of mobile communication systems mainly supported analog voice calls. Later generations introduced digital communication and mobile internet access.

Third generation networks enabled mobile data services such as email and web browsing. Fourth generation networks improved data speeds and enabled video streaming and social media services. The introduction of fifth generation networks brought new capabilities such as high-speed broadband communication, support for large numbers of devices, and ultra-reliable low-latency communication.

However, new digital technologies are now emerging that require even better network performance. These technologies consist with virtual reality, augmented reality, holographic communication, and large virtual environments known as the metaverse.

These immersive applications create digital environments such as users interact with virtual objects and other people. Users may wear head mounted displays or use specialized sensors and controllers.

In these environments, users expect the system to respond suddenly to their actions. For example, the image displayed must update quickly, when a user moves their head in a virtual reality environment. If the system responds slowly, the user may experience unsettle or sickness of motion.

Similarly, in remote robotic systems, operators may control machines located far away. If communication delays are too large, the control system may become unstable.

To support such applications, communication networks must provide extremely low latency and high reliability. Researchers introduced the concept of Extreme Ultra-Reliable Low-Latency Communication to address this need.

Another related concept is the Tactile Internet. The tactile internet allows users to interact with remote systems through touch and control signals.

This paper provides a survey of technologies and research related to these concepts. It explains how extreme URLLC and the tactile internet can support immersive applications in future networks.

The main goals of this survey are,

- explain immersive communication requirements
- discuss enabling technologies
- compare research approaches
- identify research challenges
- discuss future research directions

II. BACKGROUND CONCEPTS

A. Ultra-Reliable Low-Latency Communication

Ultra-Reliable Low-Latency Communication is one of the main service types introduced in 5G networks. URLLC is designed to support applications that require reliable communication with very small delays.

Typical URLLC requirements include,

- latency below 10 milliseconds
- reliability above 99.999

Include industrial automation, autonomous vehicles, and smart manufacturing systems are applications that require URLLC. In these applications, communication failures or delays may direct to serious problems.

For example, in a factory automation system, machines must coordinate their actions exactly. Communication delays could create machines to collide or malfunction. However, immersive digital applications require even faster communication. In XR systems, delays greater than twenty milliseconds can affect user experience. For this reason, researchers proposed the concept of Extreme URLLC.

B. Extreme URLLC

Extreme URLLC is an extension of traditional URLLC. It focuses on achieving extremely low latency and very stable

communication performance. Extreme URLLC systems aim to reduce not only average latency but also latency variation. This is important because sudden increases in delay can affect immersive applications.

For example, if a virtual reality system occasionally experiences high latency spikes, the user may notice sudden visual jumps or delays. Extreme URLLC systems attempt to provide deterministic communication performance. Deterministic communication means that communication delay remains within predictable limits.

To achieve extreme URLLC performance, several improvements are required. These include improvements in wireless transmission technologies, network architectures, and computing infrastructure.

C. Tactile Internet

The Tactile Internet refers to communication networks that support real-time transmission of touch and control signals. Traditional internet applications focus mainly on information exchange. As example, users send messages, watch videos, or download files. The tactile internet allows users to interact with remote environments in real time. This interaction includes physical actions and sensory feedback.

A typical tactile internet system consists of three main components. The first component is the user control device. This device allows the user to send commands. The second component is the communication network. The network transmits commands and feedback signals. The third component is the remote device or robot. The remote device executes actions based on user commands.

As example, a surgeon may control a robotic surgical instrument located in another city. The surgeon sends control commands through the network. The robot performs the operation and sends tactile feedback back to the surgeon. For this system to work safely, communication delay must be extremely small.

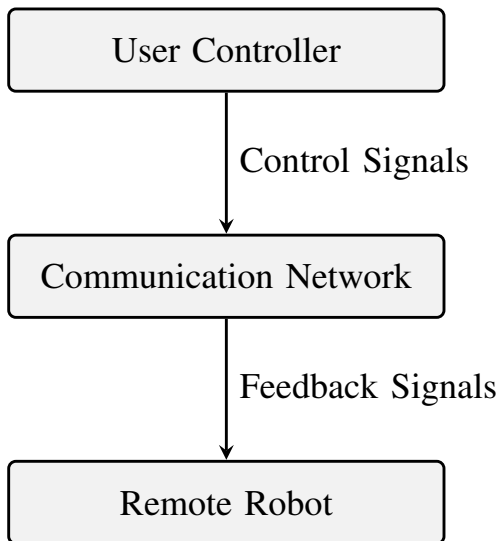


Fig. 1. Basic Tactile Internet System

III. IMMERSIVE APPLICATIONS

A. Extended Reality

Extended Reality is a common term. That includes the technologies of virtual reality, augmented reality, and mixed reality. Virtual reality creates a fully digital environment. Users wear head mounted displays to experience simulated worlds. Augmented reality covers digital objects onto the real world. As example, AR systems may display navigation information on a smartphone screen.

Mixed reality combines real elements and virtual elements. Users may interact with both physical and digital objects simultaneously. XR applications require large amounts of data processing. They also require continuous communication between user devices and computing servers. To provide smooth user experience, XR systems must operate with very low latency.

B. Holographic Communication

Holographic communication represents an advanced form of digital communication. It allows users to interact with three dimensional images of other people. Unlike traditional video calls, holographic communication creates realistic 3D representations of participants.

These holographic images may be generated using multiple cameras and sensors. However, transmitting holographic data requires extremely high bandwidth and low latency communication. Therefore holographic communication is considered one of the most demanding applications for future networks.

C. Metaverse

The metaverse is a steady virtual environment where users interact with each other using avatars. Metaverse platforms support activities such as,

- Gaming
- Social interaction
- Education
- Remote collaboration

These environments require continuous communication between users and servers. Large numbers of users may access the same environment simultaneously.

D. Communication Requirements

Immersive applications require better communication performance than traditional applications.

Important requirements include:

- ultra-low latency
- high reliability
- high bandwidth
- stable communication

Even small communication delays may affect user experience.

TABLE I
APPLICATION REQUIREMENTS

Application Type	Bandwidth Requirement	Latency Requirement
Video Streaming	Medium	100 ms
Online Gaming	High	50 ms
XR Applications	Very High	20 ms
Tactile Internet	High	1–5 ms

E. Sources of Network Delay

- Communication delay occurs due to several factors in communication systems.
- Transmission delay occurs when data travels through communication links.
- Processing delay occurs when network devices analyze and process data packets.
- Queueing delay occurs when packets wait in buffers before transmission.
- Propagation delay occurs when signals travel across physical distances.

Reducing these delays is important for supporting immersive applications.

F. Quality of Experience

Quality of Experience measures user satisfaction. High QoE is necessary for immersive applications.

Important QoE factors include,

- latency
- frame rate
- image quality
- synchronization

TABLE II
QOE METRICS

Metric	Description
Latency	System response time
Frame rate	Frames per second
Packet loss	Lost data packets
Jitter	Delay variation

IV. ENABLING TECHNOLOGIES FOR IMMERSIVE COMMUNICATION

A. Edge Computing

Edge computing is an important technology for reducing communication delay. Traditional cloud computing systems process data in centralized data centers. These data centers may be located far from users. Edge computing places computing resources close to users. Edge servers are usually located near base stations.

Edge servers may perform tasks such as,

- graphics rendering
- machine learning
- data analysis

Processing data closer to users reduces communication delay and improves system performance.

Traditional Model



Edge Model

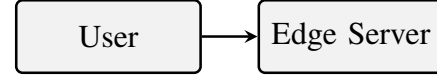


Fig. 2. Comparison of traditional cloud and edge computing architectures

B. Artificial Intelligence for Network Management

Artificial intelligence can improve network performance. AI algorithms can analyze network traffic patterns. They can predict future network conditions and adjust resource allocation accordingly. AI systems may also optimize routing decisions and reduce network congestion. Using AI for network management can help improve communication performance in immersive systems.

Benefits of AI networking include,

- improved resource allocation
- reduced latency
- better system performance

TABLE III
AI TECHNIQUES FOR NETWORKING

AI Method	Application
Neural Networks	Traffic prediction
Reinforcement Learning	Resource allocation
Deep Learning	Network anomaly detection

AI will play an important role in future immersive communication systems.

C. Network Slicing

Network slicing stands for operators to divide a physical network into multiple virtual networks. Each slice supports a specific application.

Examples,

- XR slice
- IoT slice
- video streaming slice

Network slicing allows the network to allocate resources efficiently.

D. Advanced Wireless Technologies

Future networks may use advanced wireless technologies to improve performance.

Examples include,

- terahertz communication
- massive MIMO
- intelligent reflecting surfaces

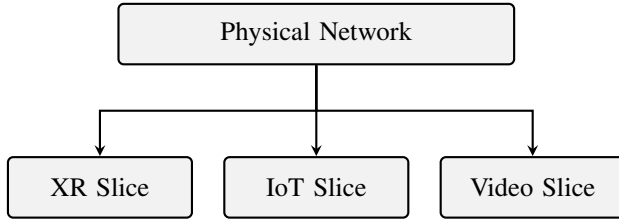


Fig. 3. Network slicing architecture

These technologies can improve network capacity and reduce communication latency.

- **Energy Efficiency** - XR devices require powerful processors and high resolution displays. These components consume large amounts of energy. Battery life is therefore an important concern. Edge computing can reduce device energy consumption by offloading heavy tasks. Energy-efficient communication protocols are also necessary.

TABLE IV
ENERGY CONSUMPTION SOURCES

Component	Energy Use
Display system	High
Wireless communication	Medium
Processing unit	High
Sensors	Low

- **Scalability Challenges** - Large scale metaverse environments may involve thousands of users. Communication networks must support these users simultaneously. Edge servers may become overloaded during peak usage. Load balancing techniques can distribute traffic across multiple servers. Scalable architectures are essential for future immersive systems. degradation.
- **Mobility Challenges** - Users of immersive applications often move while interacting with virtual environments. Network handovers occur when users move between base stations. Frequent handovers may increase communication delay. Advanced mobility management techniques are required. These techniques must maintain stable communication during user movement.

E. XR Rendering Pipeline and Communication Flow

XR systems require a complex processing pipeline. The pipeline includes sensing, data transmission, rendering, and display. The first stage of the pipeline is user sensing. Sensors in XR devices capture user movements. Examples include head movement, hand gestures, and eye tracking. The second stage is data transmission. The captured data must be transmitted to a computing system. This system may be located on the device, at the edge server, or in the cloud.

The third stage is graphics rendering. Rendering generates the visual content that the user sees in the virtual environment. The final stage is display output. The rendered image is displayed to the user through the XR headset. If any stage in this pipeline experiences delay, the user experience is affected.

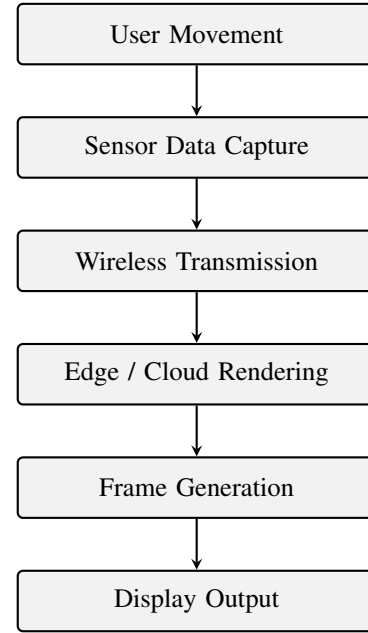


Fig. 4. XR rendering pipeline

F. End-to-End Latency Pipeline

Communication delay in immersive systems occurs across multiple components.

These components include:

- device processing
- wireless transmission
- network routing
- edge processing
- display rendering

Each stage contributes to the total delay.

TABLE V
LATENCY CONTRIBUTION OF EACH COMPONENT

System Component	Typical Delay
Sensor processing	2 ms
Wireless transmission	3 ms
Edge computation	5 ms
Rendering	5 ms
Display update	3 ms

Total delay must remain below 20 milliseconds for smooth XR experiences.

G. Metaverse Communication Architecture

Total delay must remain below 20 milliseconds for smooth XR experiences. Metaverse systems require communication between multiple components.

These components include:

- user devices
- edge servers
- cloud platforms
- digital asset servers

User devices send interaction data to edge servers. Edge servers process local interactions. Cloud servers handle large-scale world simulations.

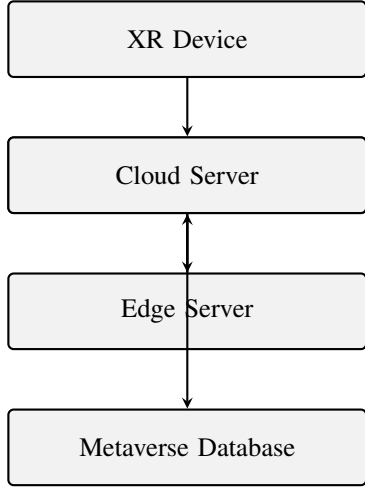


Fig. 5. Immersive Communication Architecture

H. Data Traffic Characteristics of Immersive Systems

Immersive applications generate large volumes of data. XR applications transmit,

- high resolution video frames
- motion tracking data
- environmental updates

Holographic communication produces even larger data streams. These streams include multiple camera views and depth information. Metaverse platforms generate continuous interaction data between users.

TABLE VI
DATA RATE REQUIREMENTS

Application	Data Rate
Video streaming	25 Mbps
VR applications	100 Mbps
Holographic communication	1 Gbps
Metaverse environments	200 Mbps

Future networks must support these high data rates.

V. SURVEY AND COMPARISON OF EXISTING RESEARCH

A. Research Methodologies

Researchers have proposed different approaches for supporting immersive communication systems.

Common approaches include,

- edge computing optimization
- resource allocation strategies
- network architecture design

Each approach addresses different aspects of the problem.

B. Case Study – XR Gaming

XR gaming is one of the most popular immersive applications. Users interact with virtual environments using head mounted displays and motion sensors in XR game. The system must update the virtual environment immediately when the user moves. Edge computing can reduce delay in XR gaming systems.

C. Case Study - Remote Robotics

Remote robots can perform tasks in dangerous environments. Operators control robots through communication networks. Extreme URLLC communication is required for stable robotic control. Examples include,

- underwater exploration
- industrial maintenance
- space missions

VI. RESEARCH CHALLENGES AND FUTURE DIRECTIONS

A. Role of 6G Networks

6G networks are expected to provide significant improvements over 5G.

Key features of 6G include,

- ultra-high data rates
- extremely low latency
- advanced AI-driven networking
- integration of communication and sensing

6G networks may operate at frequencies above 100 GHz. These high frequencies enable large bandwidth availability. However, high frequency signals have limited coverage. Therefore, dense network deployment is required.

B. Integrated Communication and Computing

Future networks will integrate communication and computing resources. This integration allows data processing to occur closer to users. Edge computing is a key part of this approach.

Edge servers perform tasks such as:

- XR rendering • machine learning inference • data filtering

This reduces communication delay.

C. Security Considerations

Immersive systems collect sensitive user data.

Examples include,

- location data
- biometric information
- motion tracking data

Strong security mechanisms are required. Security must not increase communication delay.

D. Future Research Directions

Future research should focus on AI based network optimization, cooperative edge computing, deterministic networking, and advanced wireless technologies. These innovations will support immersive communication systems.

TABLE VII
SUMMARY OF RELATED LITERATURE

Paper	Application	Method	Advantage	Limitation
Promwongsa et al. [1]	Tactile Internet	Architecture survey	Provides comprehensive overview of tactile internet concepts and requirements	Does not include implementation or experimental validation
Hou et al. [2]	Tactile Internet / 6G	AI-based networking	Introduces intelligent communication using AI for ultra low latency networks	Mostly conceptual framework with limited practical evaluation
Clemm et al. [3]	Holographic communication	System architecture design	Clearly defines requirements for immersive holographic communication systems	Limited experimental performance analysis
Petkova et al. [4]	Holographic communication	Technology analysis	Discusses latest technologies for holographic communication and 6G integration	Mainly theoretical discussion
Sarah et al. [5]	URLLC with MEC	Resource allocation	Proposes efficient resource allocation using mobile edge computing	Simplified network model used in evaluation
Zhu et al. [6]	Virtual Reality streaming	Edge computing offloading	Reduces latency and energy consumption using edge rendering	Requires strong edge infrastructure
Xie et al. [7]	VR streaming	Edge-based streaming optimization	Improves video delivery efficiency for VR applications	Limited scalability analysis
Choudhury et al. [8]	XR applications	Experimental network evaluation	Provides real world evaluation of XR network performance	Focused mainly on experimental environment
Gao et al. [9]	XR rendering	Rendering offload framework	Reduces device processing load and latency for XR systems	High dependency on edge computing resources
ACM Metaverse Study [10]	Metaverse networking	Network performance analysis	Studies network requirements for large scale metaverse platforms	Limited implementation level evaluation

VII. CONCLUSION

Immersive technologies such as XR, holographic communication, and the metaverse require very fast and reliable communication networks. Traditional networks are not fully capable of supporting these applications because they need extremely low latency and stable connections.

Extreme Ultra Reliable Low Latency Communication and the Tactile Internet provide important solutions for these requirements. Technologies such as edge computing, artificial intelligence, and network slicing can also improve network performance for immersive systems.

We reviewed recent research related to these technologies and also compared different approaches. Although many solutions have been proposed, some of challenges such as latency, scalability, security, and energy efficiency still remain.

Future research should focus on improving network architectures and intelligent network management to support immersive applications in future networks such as 6G.

REFERENCES

- [1] N. Promwongsa, A. Ebrahimzadeh, D. Naboulsi, S. Kianpisheh, F. Belqasmi, R. H. Glitho, N. Crespi, and O. Alfandi, "A comprehensive survey of the tactile internet: State-of-the-art and research directions," *IEEE Communications Surveys & Tutorials*, vol. 23, no. 1, pp. 472–523, 2021.
- [2] Z. Hou, C. She, Y. Li, D. Niyato, and T. Q. S. Quek, "Intelligent communications for tactile internet in 6G: Requirements, technologies, and challenges," *IEEE Communications Magazine*, vol. 59, no. 12, pp. 82–88, Dec. 2021.
- [3] A. Clemm, M. Torres Vega, H. K. Ravuri, T. Wauters, and F. De Turck, "Toward truly immersive holographic-type communication: Challenges and solutions," *IEEE Communications Magazine*, vol. 58, no. 1, pp. 93–99, Jan. 2020.
- [4] R. Petkova, G. Bozhilov, D. E. Dimitrov, P. Tsvetkov, H. Karamfilov, and M. K. Kassarov, "On the way to holographic-type communications: Perspectives and enabling technologies," *IEEE Access*, vol. 12, pp. 59236–59272, 2024.
- [5] A. Sarah et al., "Resource allocation in multi-access edge computing for URLLC services in 5G and 6G: State-of-the-art and future challenges," *Computer Networks*, vol. 227, 2023.
- [6] Z. Zhu, X. Feng, Z. Tang, and S. Wei, "Power-efficient live virtual reality streaming using edge offloading," in *Proc. ACM Workshop on Network and Operating Systems Support for Digital Audio and Video (NOSSDAV)*, 2022.
- [7] Z. Zhu, N. Jiang, T. Guo, and S. Wei, "Virtual reality streaming at the edge: A power perspective," in *Proc. ACM/IEEE Symposium on Edge Computing (SEC)*, 2019.
- [8] S. Choudhury, J. Kolodziejewski, I. Seskar, and D. Raychaudhuri, "Experimental evaluation of virtual reality applications running on next-generation network scenarios with edge cloud assistance," in *Proc. ACM International Conference on Mobile Computing and Networking (MobiCom)*, 2022.
- [9] S. Gao, J. Liu, Q. Jiang, F. Sinclair, W. Sentosa, B. Godfrey, and S. V. Adve, "XRgo: Design and evaluation of rendering offload for low-power extended reality devices," in *Proc. ACM Multimedia Systems Conference (MMSys)*, 2025.

- [10] “Can my WiFi handle the metaverse? A performance study,” in *Proc. ACM Conference on Emerging Networking Experiments and Technologies*, 2023.